

## Tutorial No : 3

**Tutorial Title:** Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap.

**Name of Student:** Pratima Badrinarayan Chandak

**Class:** ENTC

**Div:** 2

**PRN No:**B25ET1131

**Batch:**C

---

### Program:

```
#include <iostream>
using namespace std;
```

```
void read(int arr[], int n) {
    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++)
        cin >> arr[i];
}
```

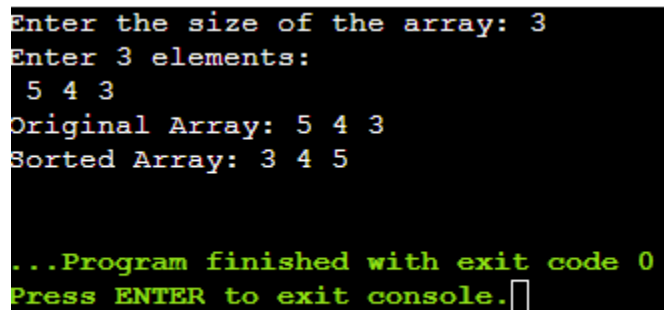
```
void display(int arr[], int n) {
    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";
    cout << endl;
}
```

```
void swap(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}
```

```
void sort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                swap(arr[j], arr[j + 1]);
            }
        }
    }
}
```

```
    }  
}  
  
int main() {  
    int n;  
  
    cout << "Enter the size of the array: ";  
    cin >> n;  
  
    int arr[n];  
  
    read(arr, n);  
  
    cout << "Original Array: ";  
    display(arr, n);  
  
    sort(arr, n);  
  
    cout << "Sorted Array: ";  
    display(arr, n);  
  
    return 0;  
}
```

### Output:

A screenshot of a terminal window with a black background and white and green text. The output shows the program's execution: it prompts for the array size (3), then for 3 elements (5, 4, 3), displays the original array (5 4 3), displays the sorted array (3 4 5), and finally shows a green message indicating the program finished with exit code 0 and prompts to press ENTER to exit the console.

```
Enter the size of the array: 3  
Enter 3 elements:  
5 4 3  
Original Array: 5 4 3  
Sorted Array: 3 4 5  
  
...Program finished with exit code 0  
Press ENTER to exit console. □
```